TIC 264099112

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        5.974399 d
  Epoch (t0)    3485.6575 BTJD
  Duration      0.1200 d
  Depth         19.834 %  (198,344 ppm)
  SNR           483.5

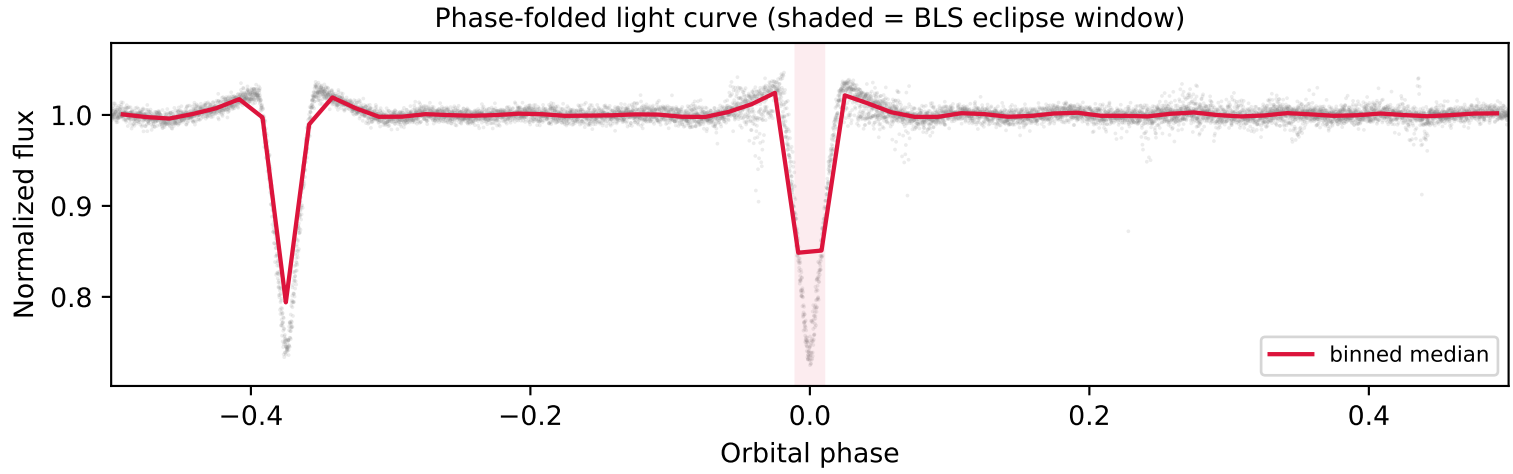
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      4  (gate >= 3)
  Robust depth ratio     1.02  (gate >= 0.50)
  Robust depth sigma     337.5  (gate >= 7.0)
  Odd/even z-score       0.23  (consistent)
  Secondary eclipse      detected, phase -0.38, 8.1 sigma
```

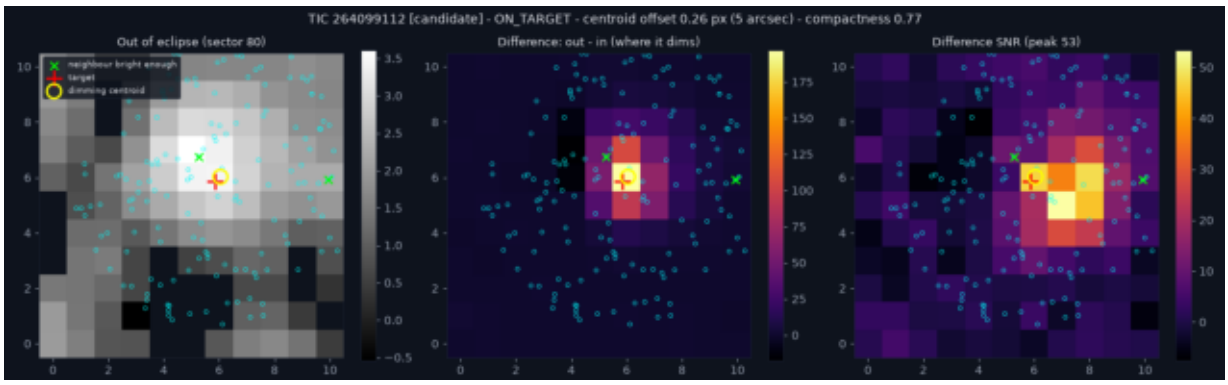
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 80    light-curve dip in this sector: 354.8 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   53.38   (gate >= 5.0)      Compactness   0.77   (gate >= 0.50)
  Dimming centroid offset     0.26 px (5 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 264099122
    1.07 px away, dTmag -1.94 - farther from the dimming than the target
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No caveats - clean detection on every check.